
Full pairwise cost
Standard QAOA
Warm-start QAOA
Multi-angle QAOA
Ideal simulation
Aachen · calibrated noise
Aachen · real device
Berlin · calibrated noise
Berlin · real device
Mean full pairwise cost
10
8
6
4
2
1
2
3
QAOA depth p
Lines: mean across repeats; small dots: individual repeats. Ideal results are sampled. Dashed: reference where applicable.